COURSE PROJECT · PROGRESS REPORT

SecondHelping

Campus Surplus-Food Marketplace

Dining hall staff post discounted "closing bags" near closing time; students browse, filter, reserve, and pick up with a code. This report covers system development, deliverables, and team collaboration.

01 Technical Progress — System Development

02 Experimental Progress — Deliverables

03 Project Progress — Team Collaboration & Management

The Debuggers · Software Engineering



AGENDA

Three parts: system, deliverables, team

01 Technical Progress — System Development

Overall system requirements, the foundation already in place, and the classification of remaining development work.

02 Experimental Progress — Deliverables

Completed parts of the requirements documentation, and the division of responsibilities for the remaining Assignment 1 work.

03 Project Progress — Team Collaboration & Management

Roles and responsibilities of team members, allocation of experimental tasks, and how the team works together.

01

Technical Progress

System Development

Overall system requirements, the existing foundation, and the classification and structure of requirements that remain to be developed.



What SecondHelping does — a brief review

THE PROBLEM

When dining halls are about to close, the staff discard the remaining unsold meals. At the same time, students on the same campus are looking for affordable food nearby.

THE SOLUTION

A campus marketplace where staff post surplus food (**closing bags**) at a discount price and students reserve them for pickup.

USERS

Students — sign up, log in, browse and filter bags, reserve a meal, show a pickup code.

Dining hall supervisors — publish closing bags, confirm pickups, track rescued meals and savings.

CORE FLOW

1. **Staff publish a closing bag**
   Quantity, regular price, discount, allergens, pickup window
2. **Students browse and filter**
   Filter available meals by dining hall and allergens
3. **Reserve a meal**
   Each reservation receives a unique pickup code
4. **Pickup at the dining hall**
   Student shows the pickup code within the pickup window
5. **Staff confirm pickup**
   Dashboard updates rescued meals, orders, and student savings

A running full-stack prototype is already in place

TECH STACK

Application
Next.js 16 (App Router) · React 19 · TypeScript · Tailwind CSS 4

Server logic
Next.js Server Actions — marketplace actions + auth actions

Data
PostgreSQL 16 via Docker Compose · Prisma 7 with committed migrations and a demo seed

Security
Hashed passwords · database-backed sessions · route guards per role

Toolchain
Node.js 20+ · pnpm · ESLint

DATA MODEL — 5 ENTITIES

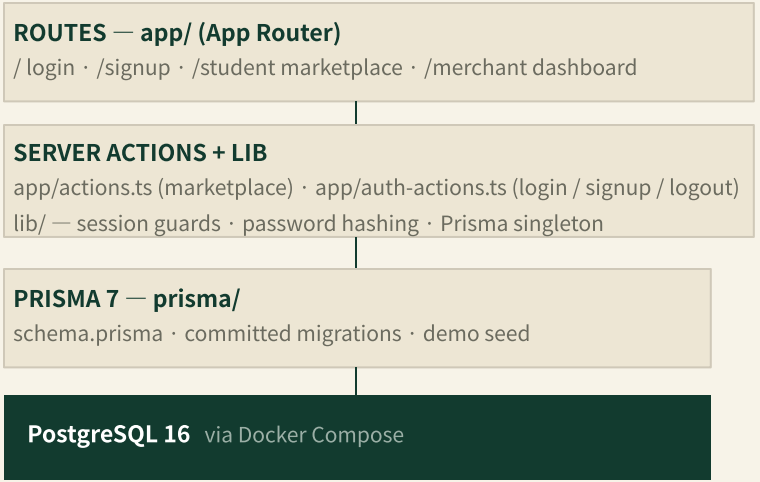
User — student / supervisor roles

AuthSession — session tokens stored as hashes

DiningHall — campus dining halls

Listing — closing bags: quantity, price, discount, allergens, pickup window

Order — reservation with pickup code



Core flows are implemented; remaining work groups into three classes

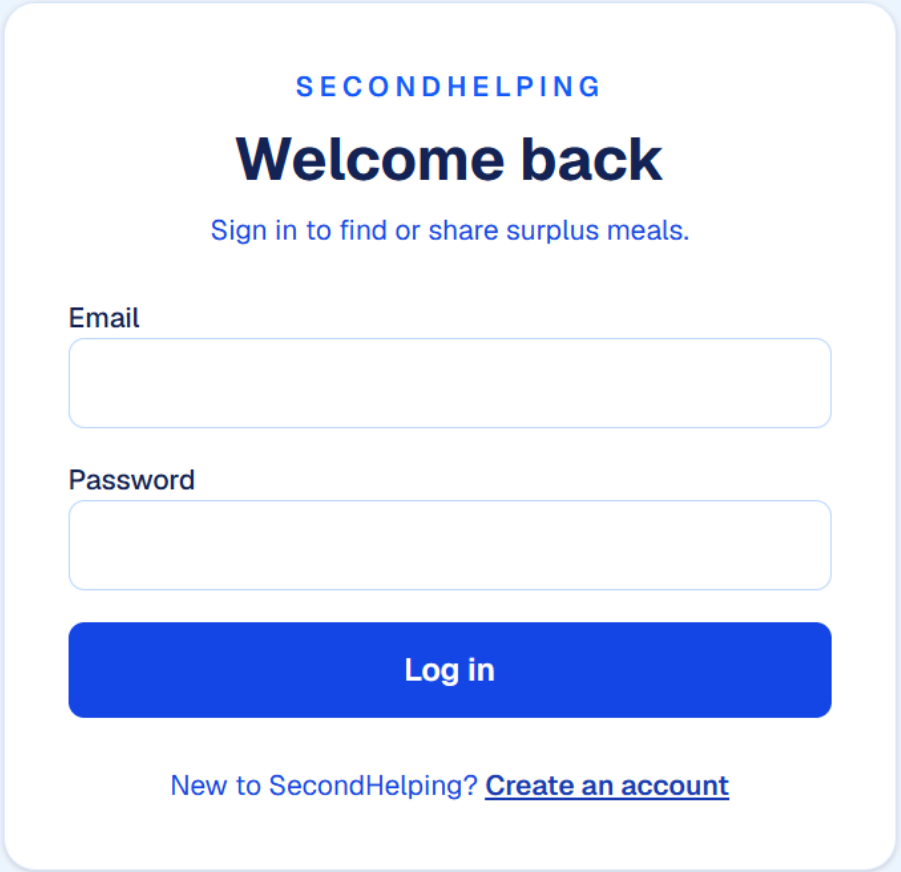
Already implemented: auth & role guards · publish closing bags · browse & filter · reservation with pickup code · pickup confirmation

Requirement class	Scope	Status
A · Product features	Online payment for reserved bags	To develop
	Notifications (reservation / pickup reminders)	To develop
	Scan-to-verify pickup codes (camera / QR)	To develop
B · Administration	Supervisor account provisioning UI (currently seeded manually)	To develop
	Listing management refinements (edit / close a bag)	To develop
C · Engineering & rollout	Automated tests for server actions and auth flows	To develop
	Deployment to a public host with managed PostgreSQL	To develop
	Security pass — rotate demo credentials before any deployment	To develop

Basis: prototype scope stated in the README — payments, notifications, and code scanning are explicitly out of scope for the current prototype.

From reactive rescue to predictive, fair campus food circulation AI MODULES

AI module	Core Function	Expected Outcome
Predictive Surplus Engine	Forecast leftover using sales, menu data, weather, campus events, schedules, and historical waste	Expected surplus + confidence intervals; early prep-reduction signals
Pre Copilot + Digital Twin	Simulate pre-level trade-offs and alternative service scenarios	Recommended prep changes; overproduction reduced before waste appears
Multimodal Listing Copilot	Convert photo or voice input into a structured listing draft	Faster listing creation; consistent description, discount, and pickup window
Dynamic Pricing Agent	Adjust discounts by time remaining, demand, inventory and fairness caps	Higher sell-through without exploitative pricing
Rescue Router	Recommend next-best destination: pantry, shelter, staff meal, another kitchen, or compost	Circular-economy routing; less unsold food discarded
Impact Estimator	Track meals rescued, kg diverted, CO2 avoided, water/kand saved, and student savings	Real-time social and environmental proof for each cycle


SECONDHELPING
Welcome back
Sign in to find or share surplus meals.
Email
Password
Log in
New to SecondHelping? Create an account

Closing bags and orders

Chicken Rice Closing Bag
ACTIVE
$5.50 · 60% off · 14/14 left · 0 orders
Pickup: 2026年10月13日 18:03–2026年10月13日 18:33
Contains: soy, sesame

Beef Noodle Bowl
ACTIVE
$7.80 · 60% off · 5/5 left · 0 orders
Pickup: 2026年10月13日 18:03–2026年10月13日 18:33
Contains: wheat, soy

Seasonal Vegetable Bowl
ACTIVE
$4.80 · 60% off · 8/8 left · 0 orders
Pickup: 2026年10月13日 18:03–2026年10月13日 18:33

SECONDHELPING
Main Dining Hall

Go to Order
18:28:56
Asia/Shanghai
Log out

Meals rescued this term	Student savings this term	Orders this term
0	$0.00	0

Post a closing bag
Set the regular price, discount, quantity, allergens, and pickup window.

Meal or bag name
Quantity available
Regular price (cents)
Discount (%)
60
Pickup starts
年 /月/日 --:--
Pickup ends
年 /月/日 --:--
Allergens, comma separated
Description and pickup instructions
Publish closing bag

02

Experimental Progress

Deliverables

Completed parts of the requirements documentation, and the division of responsibilities and task allocation for the remaining Assignment 1 work.



Requirements documentation: core chapters drafted

COMPLETED PARTS

✓ Project overview and system scope

✓ Stakeholder and role definitions — student / dining hall supervisor

✓ Functional requirements for core flows — auth, publish, browse & filter, reserve, pickup confirmation

✓ Data requirements — five entities and their relations

✓ Prototype walkthrough as supporting evidence — repository plus demo accounts

EVIDENCE IN THE REPOSITORY

Route	Purpose
/	Log in
/signup	Create a student account
/student	Browse, filter, reserve, view pickup codes
/merchant	Post bags, confirm pickups

Seeded demo accounts for both roles make every documented flow reproducible in a local environment (pnpm + Docker Compose).

Accountable owner per deliverable

Deliverable	Scope	Owner
Use-case refinement	Complete use-case descriptions and diagrams for all roles	Kevin
Non-functional requirements	Performance, security, usability, reliability requirements	Bianca
Requirements traceability	Map each requirement to prototype features and tests	Marwen
Databse	Database Design, implementation and population	Gabriel
Presentation & rehearsal	Slides, demo script, timing check and coordination	Zamo

03

Project Progress

Team Collaboration & Management

Roles and responsibilities of team members, the allocation of experimental tasks, and the team's working agreements.

Each member owns a clear area of responsibility

Member	Project role	Responsibilities	Current focus
Zamo	Team lead	Schedule and coordination, submission owner, meeting notes	Scheduling tasks
Kevin	Backend	Schema, migrations, server actions, auth and session handling	Requirement–data mapping
Bianca	Frontend	Student and merchant dashboards, filtering and reservation UX	Use-case walkthroughs
Marwen	Documentation & QA	Requirements document quality, testing, review and consistency	NFR chapter
Gabriel	Database	Responsible for database schema and design	Database and population

Next Focus: AI-Module

AI module	Core Function	Member Assigned
Predictive Surplus Engine	Forecast leftover using sales, menu data, weather, campus events, schedules, and historical waste	Zamo/Gabriel
Pre Copilot + Digital Twin	Simulate pre-level trade-offs and alternative service scenarios	Marwen/Bianca
Multimodal Listing Copilot	Convert photo or voice input into a structured listing draft	Kevin/Marwen
Dynamic Pricing Agent	Adjust discounts by time remaining, demand, inventory and fairness caps	Kevin
Rescue Router	Recommend next-best destination: pantry, shelter, staff meal, another kitchen, or compost	Gabriel/Bianca
Impact Estimator	Track meals rescued, kg diverted, CO2 avoided, water/kand saved, and student savings	Gabriel

Execution plan until Assignment 1 and Module-Delivery.

PLAN TO FINISH ASSIGNMENT 1

[Week 5] — Improve the baseline, develop AI-features and complete the SRS

- Detailed use cases and non-functional requirements
- Mapping of critical modules (Digital Twin and Predictive Engine)
- System traceability matrix

[Week 6] — Review AI-module features and SRS final touches

It's important to have the SRS finished before the features.

- Cross-review among team members
- Validation of business rules (Dynamic Pricing fairness caps)

[Week 7] — Integrate what we have about the module and submit

- Consolidation of the final SRS document and requirements verification
- Dry run of the presentation/demonstration

Preview Code | Blame 343 lines (271 loc) · 22.8 KB Raw

Software Requirements Specification

For SecondHelping

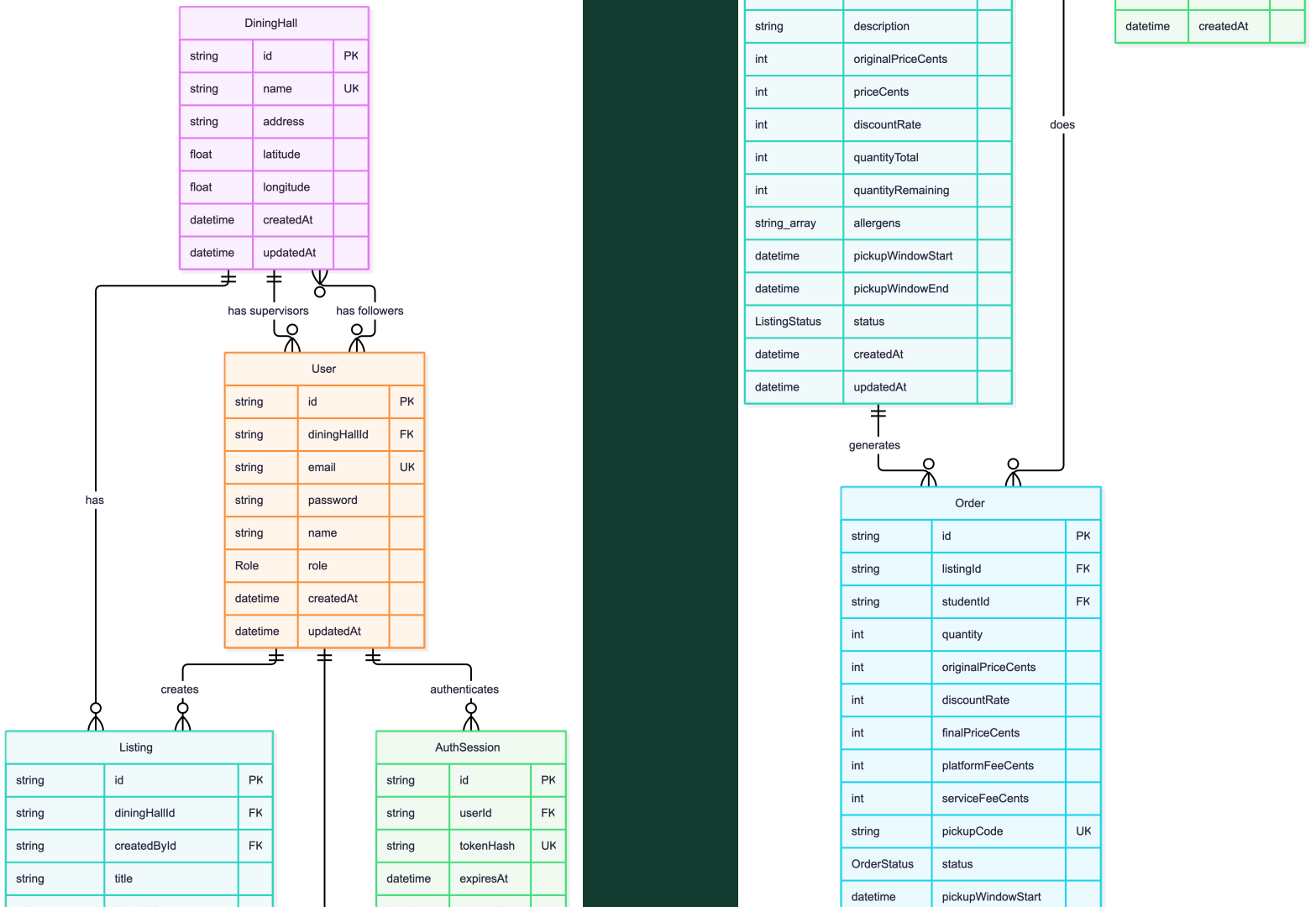
Version 0.2
Prepared by Geming Wu
Student from Beihang
2026/10/12

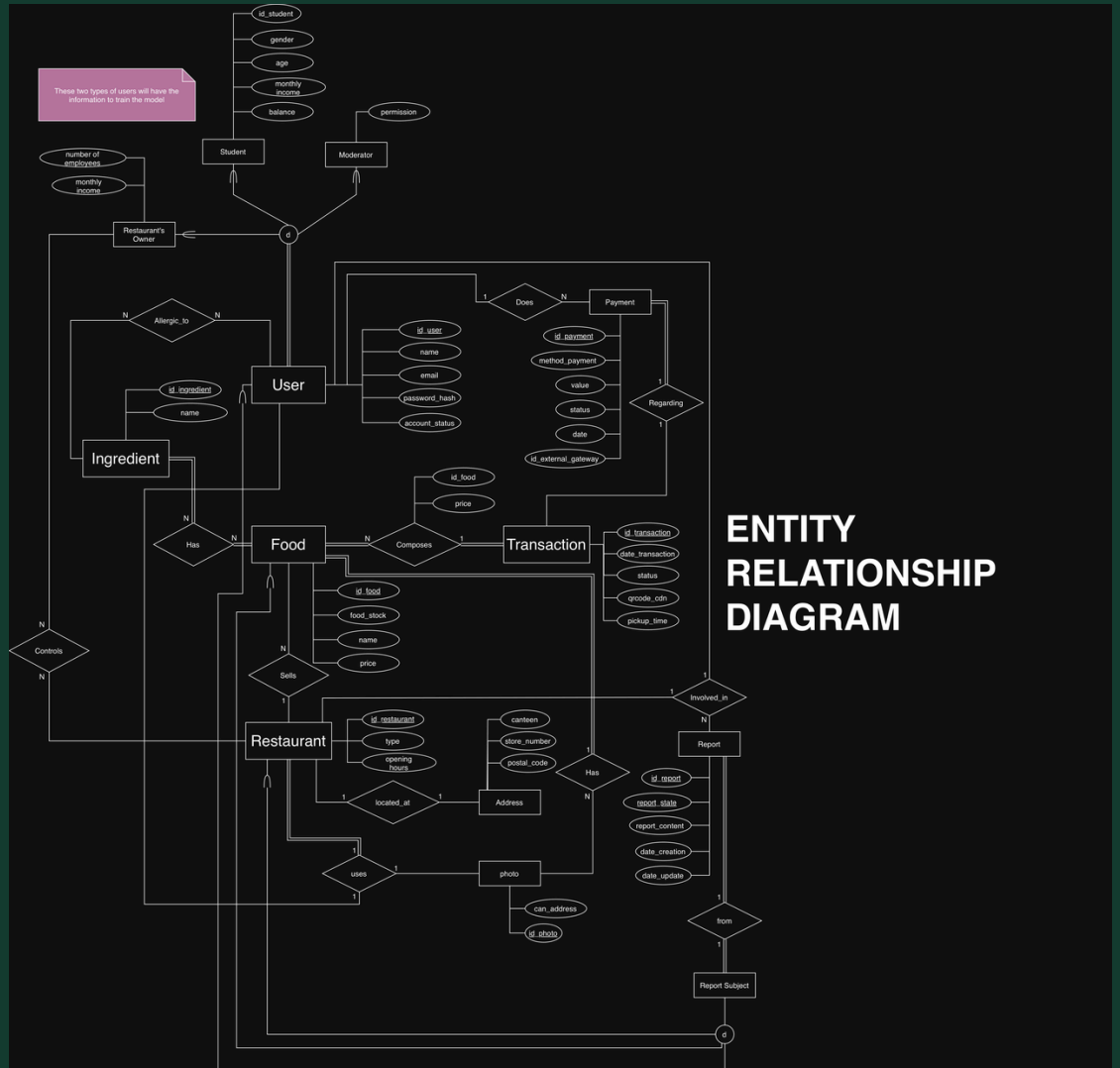
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- 1. Introduction
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  - 1.2 Product Scope
  - 1.3 Definitions, Acronyms, and Abbreviations

Thank you!
Any Questions?
Feel free to ask!




DiningHall
string id PK
string name UK
string address
float latitude
float longitude
datetime createdAt
datetime updatedAt
has supervisors
has followers
has
User
string id PK
string diningHallId FK
string email UK
string password
string name
Role role
datetime createdAt
datetime updatedAt
creates
authenticates
Listing
string id PK
string diningHallId FK
string createdById FK
string title
AuthSession
string id PK
string userId FK
string tokenHash UK
datetime expiresAt
string description
int originalPriceCents
int priceCents
int discountRate
int quantityTotal
int quantityRemaining
string_array allergens
datetime pickupWindowStart
datetime pickupWindowEnd
ListingStatus status
datetime createdAt
datetime updatedAt
generates
does
Order
string id PK
string listingId FK
string studentId FK
int quantity
int originalPriceCents
int discountRate
int finalPriceCents
int platformFeeCents
int serviceFeeCents
string pickupCode UK
OrderStatus status
datetime pickupWindowStart
datetime createdAt
datetime createdAt



ENTITY RELATIONSHIP DIAGRAM